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Phase 2b Randomized Trial Comparing Concurrent Chemoradiation With Weekly Versus 3 Weekly Cisplatin in Locally Advanced Head and Neck Squamous Cell Carcinoma

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Purpose/Objective(s): Concurrent chemoradiation is one of the standard of care in locally advanced head & neck squamous cell carcinoma (LAHNSCC). Single agent Cisplatin once in three weeks along with radiation therapy is the most widely accepted chemoradiation therapy regimen. However, weekly cisplatin provides more dose intensity and is purported to have better tolerability. The main objective of the study was to compare two chemotherapy schedules, weekly Cisplatin at 40mg/m² and three weekly Cisplatin at 80mg/m² concurrently with radiation in the treatment of LAHNSCC in terms of disease free survival and toxicity.

Materials/Methods: This study was conducted between June 2010 to March 2011 at a comprehensive cancer center in south India. Sixty six patients with LAHNSCC were randomized either to cisplatin 80mg/m² every three weeks (Arm1) or cisplatin 40mg/m² every week (Arm2) after getting an informed consent. All patients received radical radiation therapy (66 in 33#). All patients were treated with conventional radiation therapy. Acute toxicities of radiation therapy and chemotherapy were assessed weekly, using Radiotherapy and Oncology Group (RTOG) and National Cancer Institute Common Toxicity Criteria (NCI) respectively. Median follow up is twenty seven months. Disease free survival was obtained by log rank test.

Results: Thirty five patients received three weekly regimen and thirty one patients received weekly regimen. Patient characteristics were balanced in both arms. Higher percentage of patients in the weekly arm received dose of cisplatin above 200 mg/m² (71% Vs 23%; p = 0.001). There was no difference between the arms in acute toxicities except Grade II vomiting which was significantly higher in 3 weekly arm (p = 0.014). Three year DFS is 56.7 % for the three weekly arm and 59.9 % for the weekly arm (p 0.96). Survival data is pending.

Conclusions: There was no statistically significant difference in DFS between the two arms. Acute toxicities were similar in both arms except for vomiting which was higher in the three weekly regimen. Survival data is pending.

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Clinical Outcome of Eye-Conserving Treatment With Ruthenium-106 Brachytherapy for Choroidal Melanoma: Experience With the First 88 Korean Patients

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Purpose/Objective(s): Eye plaque brachytherapy is one of the standard treatment options for choroidal melanoma. In our institution, such patients were treated with ruthenium-106 (Ru-106) brachytherapy and/or additional local treatment since 2006. The aim of this study is to report early clinical outcomes of this eye-conserving treatment strategy.

Materials/Methods: Eighty eight consecutive patients who were diagnosed of choroid malignant melanoma were treated with Ru-106 brachytherapy between October, 2006 and December, 2012. In our institution, malignant melanomas of variable sizes were treated with Ru-106 brachytherapy and most of the medium to large sized tumor (height > 6 mm) received combined therapy adding local excision and/or transpupillary thermal therapy (TTT). In general, 85 Gy was prescribed to the apex of the tumor for patients without local excision, and 100 Gy to the 2-mm point from the outer surface of the sclera for patients undergoing tumor excision. Local failure was defined as enucleation/exenteration of treated eye. Follow-up time was calculated from first day of brachytherapy.

Time-to-event variables were estimated using the Kaplan-Meier method and patient groups were compared using log-rank tests. The effects of prognostic factors for survival were assessed using Cox regression.

Results: Median follow-up was 29 months. The median age of all patients was 50 (range 19-82). Median basal diameter was 11.4 mm (range 2.96-17.28 mm) and apical height was 6.79 mm (range 1.73-13.66 mm). Among the 88 patients, 13 patients eventually underwent enucleation/exenteration, resulting in a 3-year actuarial rate of local failure free survival (LFFS) of 79.7%: twelve patients underwent enucleation for local recurrence and 1 for complication. Significant prognostic factor for LFFS rate was COMS (Collaborative Ocular Melanoma Study) stage (p = .042). Distant metastases were diagnosed in 12 patients and a total of 11 patients died. The median time to metastasis was 22 months (range 12-57 months). Actuarial metastases-free survival rate was 84 % and overall survival rate was 90.3% at 3 years. The number of patients with tumor height ≥ 6 mm was fifty (57%) and combined therapy was used in the treatment of 94% (n = 47) of these patients. Three-year LFFS rate was 93.5% for tumor height < 6 mm, and 71.7% for tumor height ≥ 6 mm (p = .062). Only 1 patients with tumor size < 6 mm (n = 38) experienced distant metastasis and eventually died of disease.

Conclusions: First experience of Ru-106 for choroidal melanoma in Korea showed favorable outcome. Combined treatment may improve eye preservation rate in medium to large sized tumor. According to our knowledge, this is the first report of patients treated with brachytherapy for choroidal melanoma in Korea.

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Gross Tumor Volume Size and Oral Cavity Dose Drive Dysphagia in Definitive (Chemo) IMRT for Head and Neck Cancer

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Purpose/Objective(s): Patients with cancers of the head and neck undergoing definitive intensity modulated radiation therapy (IMRT) invariably experience some degree of dysphagia. Enteral feeding tubes (FT) are often employed in the setting of severe dysphagia, to maintain patient nutritional requirements. Persistent use of FT is a surrogate for severe dysphagia. This study examined the association between swallowing organs at risk (SWOAR) dosimetry, several clinical and radiological variables, and the duration of heavy FT (see below) use. The aim of this study was to identify factors prognostic for dysphagia to assist in the streamlining or omission of FT insertion.

Materials/Methods: This study identified 106 patients treated with definitive IMRT between 2007 and 2012. Gross disease was treated to 70 Gy in 35 fractions (7 weeks) with bilateral neck irradiation to 56 Gy. FT duration and intensity was gleaned from a prospective nutritional database to identify heavy feeding patients- defined by FT use for ≥ 75% of nutritional requirements for > 6 weeks. Differences in frequency distributions of demographics, use of chemotherapy, disease site, stage, and primary tumor volume were analyzed. SWOAR included superior, middle and inferior pharyngeal constrictors (PCM), base of tongue (BOT), esophageal inlet muscle (EIM), cervical esophagus (CE), glottis (GL), supraglottis (SGL) and cricopharyngeus muscle (CPM)- based on UMCG delineation guidelines. Oral cavity (OC) was also defined. Differences in frequency distributions of dose mean (Dmean) above and below 50 Gy were analyzed for all SWOAR and OC.

Results: Primary tumor size was associated with acute (FT>6 weeks) dysphagia (T-stage ≥3: 25.5% v 52.7%, p<0.01; GTVprimary >20 cc, 31.4% v 67.3%, p<0.01; GTVprimary >40 cc, 11.8% v 34.5%, p<0.01). Concurrent chemotherapy (45.1% v 67.3%, p = 0.03) and OC Dmean <50 Gy (11.8% v 30.9%, p = 0.02) were also significant discriminators.